

AI Engineer Foundation

Python for Data Science & Mathematics for AI

Mini-Project	Analyzing customer data, building mathematical insights, and processing data to support the AI pipeline
Time	2 weeks
Start	5 May 2026
End	15 May 2026

Dataset: Telco Customer Churn

url: [Telco Customer Churn](#)

Dataset Description

This dataset contains customer-level information from a telecommunications company, where each row represents a unique customer and each column corresponds to a specific attribute described in the metadata. It is designed for analyzing customer behavior and predicting churn.

The dataset includes the following key information: customers who have left the service within the last month, indicated by the target variable *Churn*; details about subscribed services such as phone service, multiple lines, internet service, online security, online backup, device protection, technical support, and streaming TV or movies; customer account information including tenure, contract type, payment method, paperless billing status, monthly charges, and total charges; as well as demographic characteristics such as gender, age group, and whether the customer has a partner or dependents.

This dataset is commonly used to explore customer churn prediction models and to gain practical insights into machine learning applications in business analytics. An updated version of this dataset is also available from IBM for further study and experimentation.

Questions

MODULE 1: NUMPY (10 Questions)

Q1. Load the dataset into a NumPy array using numerical features only (tenure, MonthlyCharges, TotalCharges). Check the shape and dtype of the array.

Q2. Compute the Mean, Median, and Standard Deviation for the columns tenure, MonthlyCharges, and TotalCharges using NumPy functions.

Q3. Apply Min-Max normalization to scale all numerical features to the range [0,1] using vectorized operations.

Q4. Use boolean indexing to filter customers with tenure > 12 and MonthlyCharges greater than the dataset average.

Q5. Compute the correlation matrix between numerical features using NumPy (without using Pandas).

Vectorization & Linear Algebra

Q6. Implement a fully vectorized function to compute pairwise Euclidean distance between customers using numerical features only (avoid explicit Python loops).

Q7. Implement both Min-Max scaling and Z-score normalization from scratch using NumPy and compare their results.

Q8. Compute cosine similarity between two customer vectors selected from the dataset (numerical features only).

Q9. Perform PCA manually using NumPy: construct the covariance matrix from numerical features, compute eigenvalues and eigenvectors, and project the data onto the top 2 principal components.

Q10. Implement batch processing by splitting the NumPy array into equal-sized batches and computing the mean vector for each batch.

MODULE 2: MATHEMATICS FOR AI (10 Questions)

Q1. Represent the dataset as a numerical matrix X (use numerical features only) and determine its shape and rank.

Q2. Encode the target variable Churn (Yes=1, No=0), then apply Linear Regression using tenure and MonthlyCharges to predict churn based on the model: $\hat{y} = Xw + b$

Q3. Compute the probabilities $P(\text{Churn} = 1)$ and $P(\text{Churn} = 1 | \text{Contract} = \text{Month-to-month})$ from the dataset.

Q4. Given the Mean Squared Error loss, derive and compute the gradient with respect to w for the dataset:

$$L = \frac{1}{n} \sum (y - \hat{y})^2$$

- Q5.** Compute the variance of MonthlyCharges and the covariance between MonthlyCharges and tenure.
- Q6.** Construct the covariance matrix using numerical features and compute its eigenvalues and eigenvectors.
- Q7.** Perform Singular Value Decomposition (SVD) on matrix X (numerical features only) and interpret the meaning of U , Σ , and V^T in the context of the dataset: $X = U\Sigma V^T$
- Q8.** Implement Gradient Descent to optimize the Linear Regression model parameters and visualize the loss convergence over iterations.
- Q9.** Discretize MonthlyCharges into bins (e.g., Low/High), then apply Bayes' Theorem to compute $P(\text{Churn}|\text{MonthlyCharges bin})$.
- Q10.** Apply both L1 (Lasso) and L2 (Ridge) regularization to the Linear Regression model and compare their effects on model coefficients and performance.

MODULE 3: PANDAS (10 Questions)

- Q1.** Load the dataset into a Pandas DataFrame and display head(), info(), and describe() to understand its structure.
- Q2.** Clean the dataset by handling missing or invalid values in TotalCharges and ensure correct data types for all columns.
- Q3.** Group the data by Contract and compute both the total number of customers and the churn rate for each group.
- Q4.** Create a new feature $\text{AvgCharge} = \text{TotalCharges} / \text{tenure}$ and handle potential division by zero cases.
- Q5.** Visualize the distribution of MonthlyCharges using a histogram and compare churn counts using a bar chart.
- Q6.** Create a pivot table with Contract as index, Churn as columns, and customer count as values.
- Q7.** Create tenure groups (e.g., 0–12, 12–24, 24+ months) and analyze churn rate across these groups.
- Q8.** Create a new categorical feature based on MonthlyCharges (Low, Medium, High) using appropriate binning, and analyze churn behavior across segments.
- Q9.** Generate advanced visualizations: a correlation heatmap for numerical features and a boxplot of MonthlyCharges grouped by Churn.

Q10. Perform feature engineering by encoding categorical variables (e.g., One-Hot Encoding) and prepare a clean dataset suitable for machine learning models.

Submission Requirements and Implementation Guidelines

Students are required to complete the mini-project using Python, implemented either in LMS Quanskill, VS Code (local environment), or Google Colab (cloud environment). The submission must demonstrate a complete and reproducible workflow from data loading to analysis and results.

Each submission should include: a well-structured Python script or notebook (.py or .ipynb) with clear organization of code into logical sections (e.g., data loading, preprocessing, analysis, visualization); appropriate use of libraries such as NumPy, Pandas, and Matplotlib/Seaborn; and meaningful comments explaining key steps and decisions.

Students must ensure that the code runs without errors and produces all required outputs (tables, statistics, and visualizations). In addition, a brief report (or markdown cells in Colab) should summarize the main findings, insights, and conclusions derived from the dataset.

All files should be submitted in a single package, including the source code and any necessary supporting files or instructions to reproduce the results.

Example:

telco-churn-project/

```
|
|— data/
|   |— raw/
|       |— telco_customer_churn.csv
|   |— processed/
|
|— notebooks/
|   |— analysis.ipynb
|
|— src/
```

```
| |— numpy_tasks.py
| |— math_tasks.py
| |— pandas_tasks.py
| |— utils.py
|
|— outputs/
| |— figures/
| |— tables/
| |— results.txt
|
|— report/
| |— final_report.pdf
|
|— requirements.txt
|— README.md
```

